EXPERIMENT 2

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

CSV Data Preprocessing and Visualization

Aim: Load, clean, summarize and visualize sales data.

Python Program

```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv('sales-data.csv')
print(df.head(),df.isnull().sum())
df['Sales']=df['Sales'].fillna(df['Sales'].median())
df=df.dropna(subset=['Product','Quantity','Region'])
print(df.describe())
s=df.groupby('Product')[['Sales','Quantity']].sum()
print(s)
s['Sales'].plot(kind='bar'); plt.show()
df['Date']=pd.to_datetime(df['Date'])
df.groupby('Date')['Sales'].sum().plot(); plt.show()
```

Result

The cleaned dataset, descriptive statistics, product totals and sales trend are displayed.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.